

AC Power Analysis

- Instantaneous time dependent
- Average power in u

Instantaneous and Average Power (1)

- The instantaneous power, $p(t)$:

$$p(t) = v(t) i(t) = V_m I_m \cos(\omega t + \theta_v) \cos(\omega t + \theta_i)$$

For Math

$$= \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) + \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i)$$

constant power Sinusoidal power at $2\omega t$

Now,

$$\left. \begin{aligned} v(t) &= V_m \cos(\omega t + \theta_v) \\ i(t) &= I_m \cos(\omega t + \theta_i) \end{aligned} \right\}$$

- The average power, P , is the average of the instantaneous power over one period,

$$P = \frac{1}{T} \int_0^T p(t) dt = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

θ_v = Voltage Angle

θ_i = Current "

Power Factor

- P is not time dependent
- When $\theta_v = \theta_i$, it's purely resistive load case
- When $\theta_v - \theta_i = \pm 90^\circ$, it is a purely reactive load case
 → inductive capacitive
- $P = 0$ means that the circuit absorbs no average power.

II Instantaneous and Average Power (5)

Q * Calculate the instantaneous power and average power absorbed by a passive linear network if :

$$v(t) = 80 \cos(10t + 20^\circ)$$

$$i(t) = 15 \sin(10t + 60^\circ)$$

⇒ At first Given,

$$v(t) = 80 \cos(10t + 20^\circ)$$

$$i(t) = 15 \sin(10t + 60^\circ)$$

$$= 15 \cos(10t + 60^\circ - 90^\circ)$$

$$= 15 \cos(10t + 30^\circ)$$

We know that,

instantaneously power,

$$P(t) = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) + \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i)$$

$$= \frac{1}{2} \times 80 \times 15 \cos(20 + 30) + \frac{1}{2} \times 80 \times 15 \cos(2 \times 10t + 20 - 30)$$

$$= 600 \cos(50^\circ) + 600 \cos(20t - 10^\circ)$$

$$= 385.7 + 600 \cos(20t - 10) \text{ W}$$

then, Average Power,

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

$$= \frac{1}{2} \times 80 \times 15 \cos(20 + 30)$$

$$= 385.7 \text{ W}$$

□ Instantaneous and Average Power (4)

Q * A current $I = 10 \angle 30^\circ$ flows through an impedance $Z = 20 \angle -22^\circ \Omega$. Find the average power delivered to the impedance.

⇒ At first,

$$V = IZ$$

$$= (10 \angle 30^\circ) (20 \angle -22^\circ)$$

$$= 200 \angle 8^\circ$$

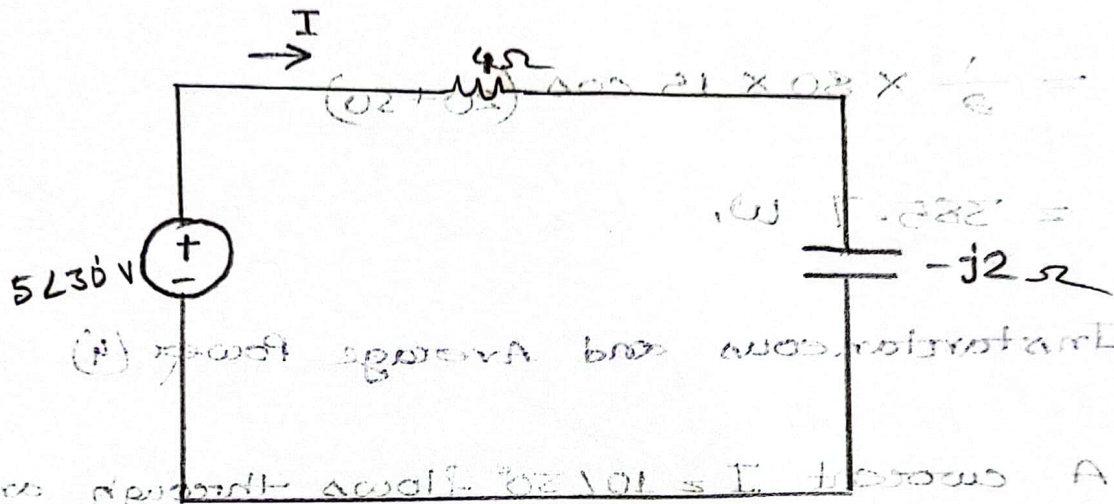
∴ the average power,

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

$$= \frac{1}{2} \times 200 \times 10 \cos(8 - 30) = 1000 \cos(-22) = 927.18 \text{ W}$$

Imp

* Example: 11.3 For the circuit shown in Fig. 11.3, find the average power supplied by the source and the average power absorbed by the resistor.



⇒ The current I is given by,

$$I = \frac{5\angle 30^\circ}{4 - j2}$$

$$= \frac{5\angle 30^\circ}{4.472 \angle -26.57^\circ}$$

$$= 1.118 \angle 56.57^\circ \text{ A}$$

The average power supplied by the voltage source is

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

$$= \frac{1}{2} \times (5) \times (1.118) \cos(30 - 56.57)$$

$$= 2.5 \text{ W}$$

The current through the resistor is

$$I_R = I = 1.118 \angle 56.57^\circ \text{ A}$$

and the voltage across it is,

$$V_R = 4I_R$$

$$= 4.472 \angle 56.57^\circ \text{ V}$$

The average power absorbed by the resistor is,

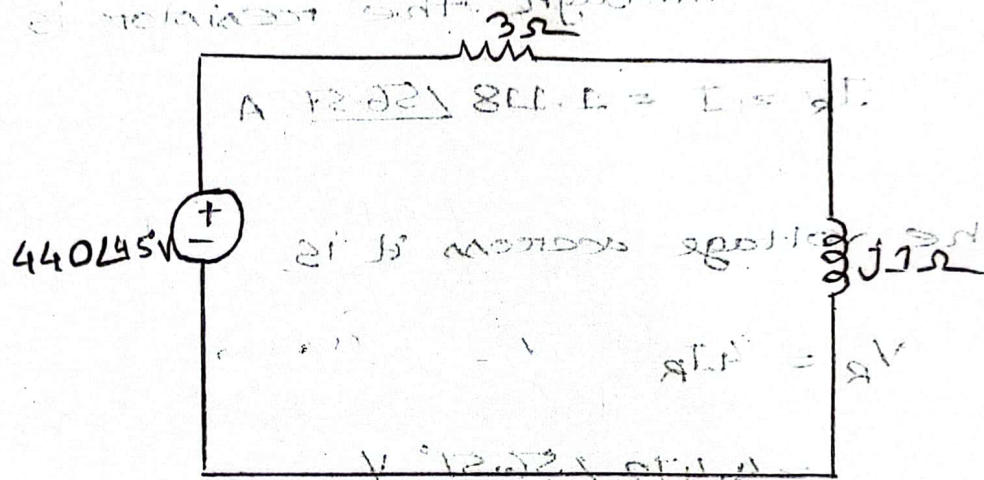
$$P = \frac{1}{2} (4.472)(1.118)$$

$$= 2.5 \text{ W}$$

Which is the same as the average power supplied

Zero average power is absorbed by the capacitor.

9 Practice Problem 11.3: In the circuit of Fig 11.3, calculate the average power absorbed by the resistor and inductor. Find the average power supplied by the voltage source.



⇒ The current I is given,

$$I = \frac{(440 \angle 45^\circ)}{3 - j1} = 4 \angle$$

$$= 139.14 \angle 63.43^\circ \text{ A [Using Calculator]}$$

Then, average power (voltage source)

$$P = \frac{1}{2} I_m V_m \cos(\theta_v - \theta_i)$$

$$= \frac{1}{2} \times 139.14 \times 440 \cos(45 - 63.43)$$

$$= 29.04 \times 10^3 \text{ W}$$

$$= 29.04 \text{ kW.}$$

then,

$$V_R = 3 \times I_R$$

$$= 3 \times 139.14 \angle 63.43^\circ$$

$$= 417.42 \angle 63.43^\circ \text{ V}$$

Now, average power, (resistor)

$$P_R = \frac{1}{2} \times V_R \times I_R \times \cos(63.43^\circ - 63.43^\circ)$$

$$= \frac{1}{2} \times 139.14 \times 417.42$$

$$= 29.04 \text{ kW.}$$

Required power is delivered in both cases.

Lastly, average power (inductor)

average power (inductor)

$$P_L = 0 \text{ W.} \quad \text{[Average power inductor is zero]}$$

for inductor, average power is zero.

□ Effective or RMS value (2)

$$I_{rms} = \frac{I_m}{\sqrt{2}}$$

[previous part-01-11 prove আছে মো]

• The average power can be written in terms of the rms values,

$$P_{avg} = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = V_{rms} I_{rms} \cos(\theta_v - \theta_i)$$

□ Apparent Power and power factor (1)

⇒ **Apparent Power**, S , is the product of the rms values of voltage and current.

⇒ It's measured in volt-ampere or VA to distinguish it from the average or real power which is measured in watts.

$$P = V_{rms} I_{rms} \cos(\theta_v - \theta_i) = S \cos(\theta_v - \theta_i)$$

Apparent power power factor, PF

⇒ **Power factor** is the cosine of the phase difference between the voltage and current.

11.5 Complex Power (1)

• Complex power S is the product of the voltage and the complex conjugate of the current:

$$V = V_m \angle \theta_v \quad I = I_m \angle \theta_i$$

$$\frac{1}{2} V I^* = V_{rms} I_{rms} \angle (\theta_v - \theta_i)$$

$$\cos(\theta_v - \theta_i) = \cos \phi \quad \text{Power factor}$$

$$\Rightarrow S = V_{rms} I_{rms} \cos(\theta_v - \theta_i) + j V_{rms} I_{rms} \sin(\theta_v - \theta_i)$$

$$S = P + jQ$$

P : is the average power in watts delivered to a load and it is the only useful power.

Q : is the reactive power exchange between the source and the reactive part of the load. It is measured in VAR.

For $Q \geq 0$ for resistive loads (unity pf)

• $Q < 0$ for capacitive (leading pf)

• $Q > 0$ inductive (lagging pf)

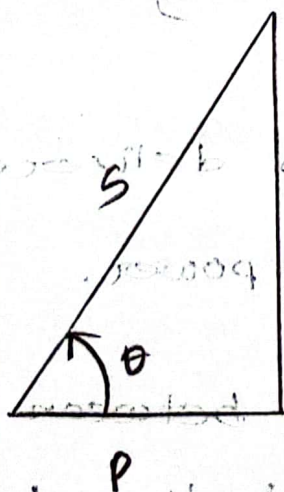
* Apparent Power, $S = |S| = V_{rms} \cdot I_{rms} = \sqrt{P^2 + Q^2}$

* Real Power, $P = \text{Re}(S) = S \cos(\theta_v - \theta_i)$

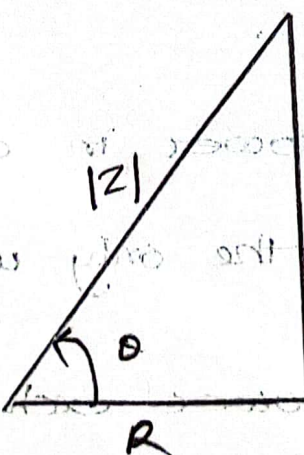
* Reactive power, $Q = \text{Im}(S) = S \sin(\theta_v - \theta_i)$

* Power factor, $\text{pf} = P/S = \cos(\theta_v - \theta_i)$

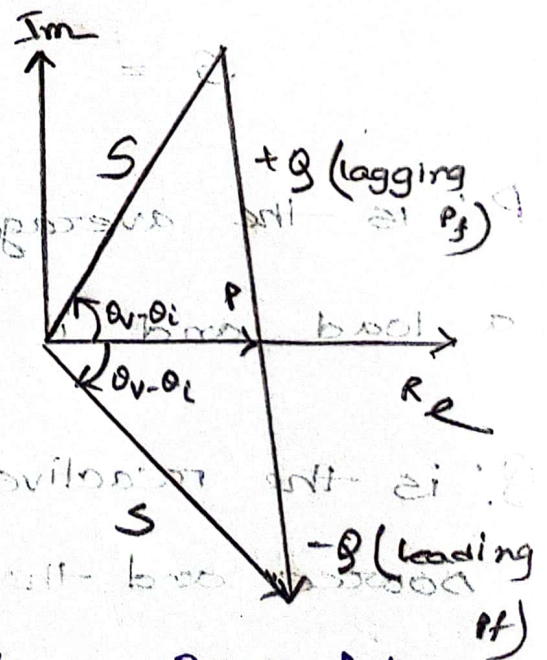
□ Complex Power (4)



power triangle



Impedance Triangle



Power factor

* Example: 11.11 The voltage across a load is $v(t) = 60 \cos(\omega t - 10^\circ) \text{ V}$ and the current through the element in the direction of the voltage drop $i(t) = 1.5 \cos(\omega t + 50^\circ) \text{ A}$, Find:

- the complex and Apparent Powers
- real and reactive power
- power factor and load impedance

⇒ a. for the rms values of the voltage and current, we write,

$$V_{\text{rms}} = \frac{60}{\sqrt{2}} \angle -10^\circ \quad I_{\text{rms}} = \frac{1.5}{\sqrt{2}} \angle +50^\circ$$

The complex power is

$$S = V_{\text{rms}} I_{\text{rms}}^* = \left(\frac{60}{\sqrt{2}} \angle -10^\circ \right) \left(\frac{1.5}{\sqrt{2}} \angle -50^\circ \right)$$

$$= 45 \angle -60^\circ \text{ VA}$$

(Question is
the sign
is opposite
sign)

The apparent power is

$$S = |S| = 45 \text{ VA}$$

b. We can express the complex power rectangular form as

$$S = 45 \angle -60^\circ$$

$$= 45 [\cos(-60^\circ) + j \sin(-60^\circ)]$$

$$= 22.5 - j38.97$$

Since,

$S = P + jQ$, the real power is

$$P = 22.5 \text{ W}$$

while the reactive power is

$$P = -38.97 \text{ VAR}$$

c. The power factor is,

$$\text{pf} = \cos(-60^\circ)$$

$$= 0.5 \text{ (leading)}$$

It's leading, because the reactive power is negative. The load impedance is,

$$Z = \frac{V}{I} = \frac{60 \angle -10^\circ}{1.5 \angle 50^\circ} = 40 \angle -60^\circ \Omega$$

which is a capacitive impedance.

* Practice Problem 11.12 | For a load, $V_{rms} = 110 \angle 85^\circ \text{ V}$,

$$I_{rms} = 3 \angle 15^\circ \text{ A}$$

Determine:

a. the complex and apparent powers

b. the real and reactive

c. the power factor and the load impedance.

A) a. $S = V_{rms} I_{rms}^*$

$$= 110 \angle 85^\circ \times 3 \angle -15^\circ$$

$$= 330 \angle 70^\circ \text{ VA} \quad [\text{complex powers}]$$

then apparent powers,

$$S = |S| = |330| = 330 \text{ VA}$$

b. $S = P + jQ$ I know that,

Now, $S = 330 \angle 70^\circ$

$$= (112.87 + j310.09) \text{ V}$$

∴ the real Power is,

$$P = 112.87 \text{ W}$$

while the reactive power is,

$$P = 310.09 \text{ VAR}$$

$$\text{C. p.f} = \cos(70^\circ)$$

$$= 0.342 \text{ (lagging)}$$

Now,

$$Z = \frac{V}{I} = \frac{110 \angle 85^\circ}{3 \angle 15^\circ}$$

$$= 36.67 \angle 70^\circ$$

$$= (12.541 + j34.46) \Omega$$

$$|Z| = 36.67$$

$$P = 112.87 \text{ W}$$

$$Q = 310.09 \text{ VAR}$$

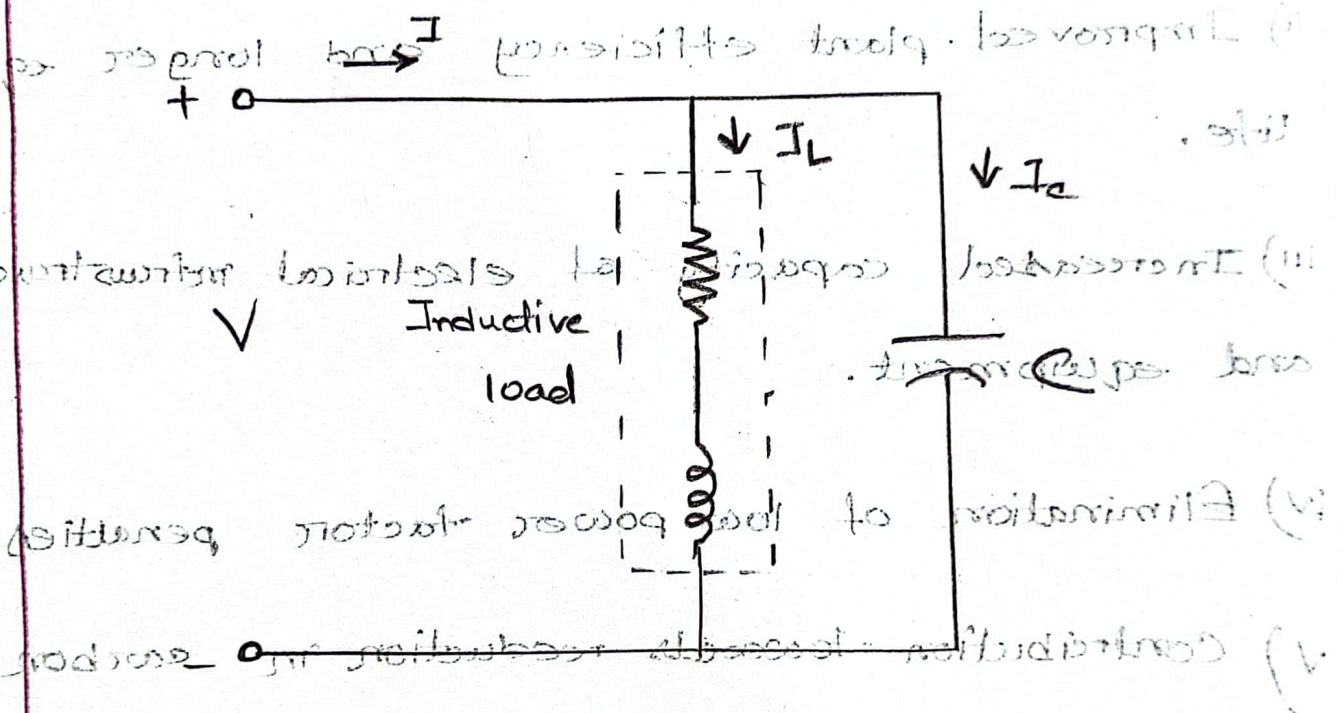
$$V = 110 \text{ V}$$

□ Advantages of Power Factor Correction: || ||

- i) Immediate, and significant savings on your electricity,
- ii) Improved plant efficiency and longer equipment life,
- iii) Increased capacity of electrical infrastructure and equipment,
- iv) Elimination of low power factor penalties
- v) Contribution towards reduction in carbon footprint.

□ Power factor correction (1)

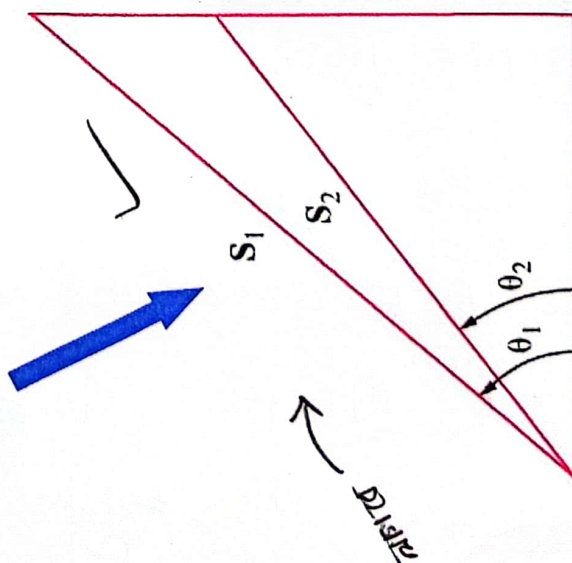
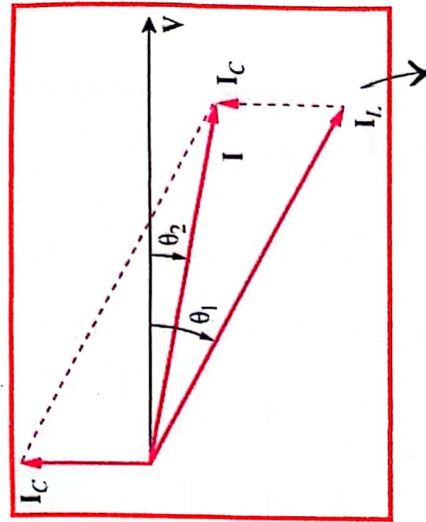
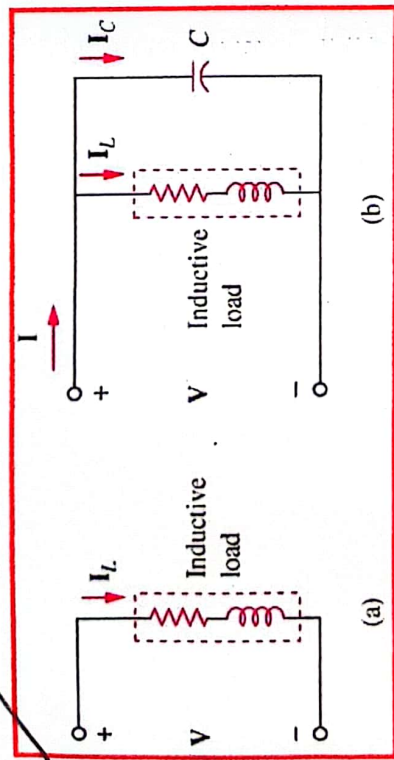
Power factor correction is the process of increasing the power factor without altering the voltage or current to the original load.



Power factor correction is necessary

for economic reason.

11.7 Power Factor Correction (2)



$$Q_c = Q_1 - Q_2$$

$$= P (\tan \theta_1 - \tan \theta_2)$$

$$= \omega C V_{2rms}^2$$

$$C = \frac{Q_c}{\omega V_{rms}^2} = \frac{P (\tan \theta_1 - \tan \theta_2)}{\omega V_{rms}^2}$$

$$Q_1 = S_1 \sin \theta_1$$

$$= P \tan \theta_1$$

$$Q_2 = P \tan \theta_2$$

$$P = S_1 \cos \theta_1$$

~~Imp~~

Practice Problem: 11.15 | Find the value of the capacitance needed to correct a load of 140 kVAR at 0.85 lagging pf to unit pf. The load is supplied by a 110 Volt (rms), 60 Hz line.

$$\Rightarrow \text{pf} = 0.85 = \cos \theta$$

$$\therefore \theta = 31.79^\circ$$

$$Q = S \sin \theta$$

$$\therefore S = \frac{Q}{\sin \theta}$$

$$= \frac{140}{\sin(31.79^\circ)}$$

$$= 265.8 \text{ kVA}$$

$$\text{For } \text{pf} = 1 = \cos \theta_1$$

$$\therefore \theta_1 = 0^\circ$$

Since P remains the same,

$$P = P_1 = S \cos \theta_1$$

$$\therefore S_1 = \frac{P_1}{\cos \theta_1} = 225.93$$

$$Q_1 = S_1 \sin \theta_1 = 0$$

The difference between the new Q_1 and the old Q is Q_c ,

$$Q_c = 140 \text{ KVAR [Given]}$$

$$= \omega C V_{rms}^2$$

$$\therefore C = \frac{140 \times 10^3}{(2\pi)(60)(110)^2} \rightarrow C = \frac{Q_c}{2\pi f \times V^2}$$

$$= 30.69 \text{ mF}$$

The End
